



EAST WEST UNIVERSITY

Assignment 01

Course Name:

Algorithm (CSE303)

Title: Analyzing the factors of laptop pricing

Section: 04

Group : D

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Submitted To:

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Analyzing the Factors of Laptop Pricing

problem statement:

In the modern digital era, laptops play a crucial role in education, professional work, and personal use. Despite their widespread use, laptop prices can vary greatly based on factors such as brand, hardware configuration, and physical design. This project seeks to explore and analyze a dataset of laptops to identify the key attributes that most significantly impact their pricing. By understanding these influencing factors, consumers and retailers alike can make more informed decisions regarding value and performance.

Over view:

This individual project focuses on performing Exploratory Data Analysis (EDA) on a dataset related to laptop pricing. The objective is to gain insights into the structure and characteristics of the data by examining variable types, detecting any missing values, analyzing feature correlations, transforming categorical variables, and uncovering pricing patterns through visualizations. The ultimate goal is to understand the key factors that influence laptop prices.

Dataset Analysis

- I. **Brand:** The manufacturer of the laptop (Categorical)
- II. **Model:** Specific model name or identifier (Categorical)
- III. **RAM (GB):** Amount of RAM in gigabytes (Quantitative – Discrete)
- IV. **Storage (GB):** Total storage capacity in gigabytes (Quantitative – Discrete)
- V. **CPU:** Type of processor used (Categorical)
- VI. **CPU_Frequency (GHz):** Processor clock speed in gigahertz (Quantitative – Continuous)
- VII. **GPU:** Graphics processing unit or graphics card model (Categorical)
- VIII. **OS:** Operating system installed on the laptop (Categorical)
- IX. **Inches:** Display size in inches (Quantitative – Continuous)
- X. **Weight (kg):** Physical weight of the laptop in kilograms (Quantitative – Continuous)
- XI. **Price (Euro):** Retail price of the laptop in Euros (Quantitative – Continuous)

Missing Values Analysis

Upon examining the dataset, we found that there are no missing values in any of the columns. As a result, no data imputation or removal of rows or columns is required, allowing us to proceed directly with analysis and modeling.

Balanced or Imbalanced Dataset

To assess the balance of the dataset, we examined the distribution of key categorical features such as **Brand** and **Operating System (OS)**. While certain brands like *Lenovo* and *HP* are more prevalent, this slight imbalance is not a concern, as the project focuses on **predicting price**—a regression task—rather than classification. Therefore, uniform distribution across categories is not essential for our analysis.

Data Types

- I. **Qualitative (Categorical):**
- II. *Brand*: Laptop manufacturer
- III. *Model*: Laptop model name
- IV. *CPU*: Type of processor
- V. *GPU*: Graphics card
- VI. *OS*: Operating system
- VII. **Quantitative (Numerical):**
- VIII. *RAM (GB)*: Amount of memory
- IX. *Storage (GB)*: Storage capacity
- X. *CPU_Frequency (GHz)*: Processor speed
- XI. *Inches*: Screen size
- XII. *Weight (kg)*: Laptop weight
- XIII. *Price (Euro)*: Retail price
- XIV. **Discrete Numerical Variables:**
- XV. *RAM (GB)*
- XVI. *Storage (GB)*
- XVII. **Continuous Numerical Variables:**
- XVIII. *CPU_Frequency (GHz)*

Descriptive Statistics

To better understand the distribution of numerical variables in the dataset, we computed key statistical measures including the **mean**, **median**, **variance**, and **standard deviation**.

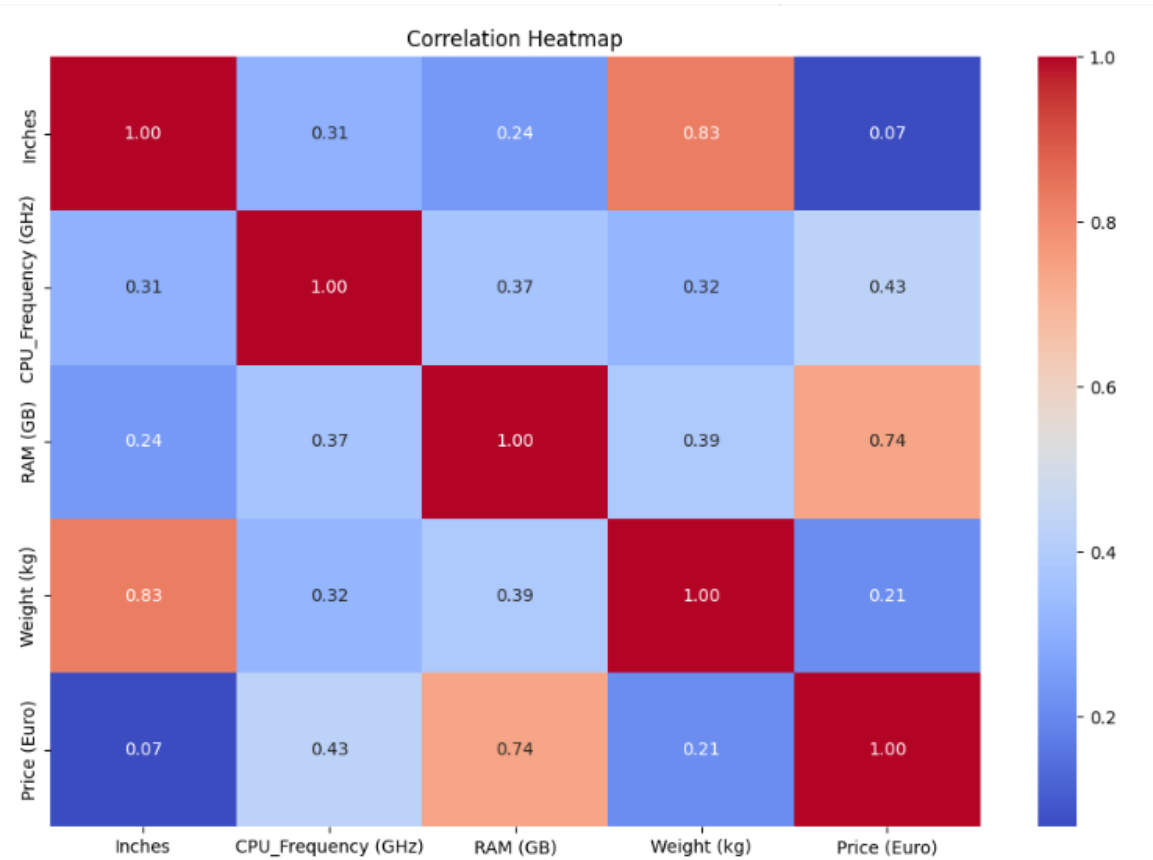
Example – Screen Size (Inches):

- **Mean:** 15.02
- **Median:** 15.60
- **Variance:** 2.04
- **Standard Deviation:** 1.43

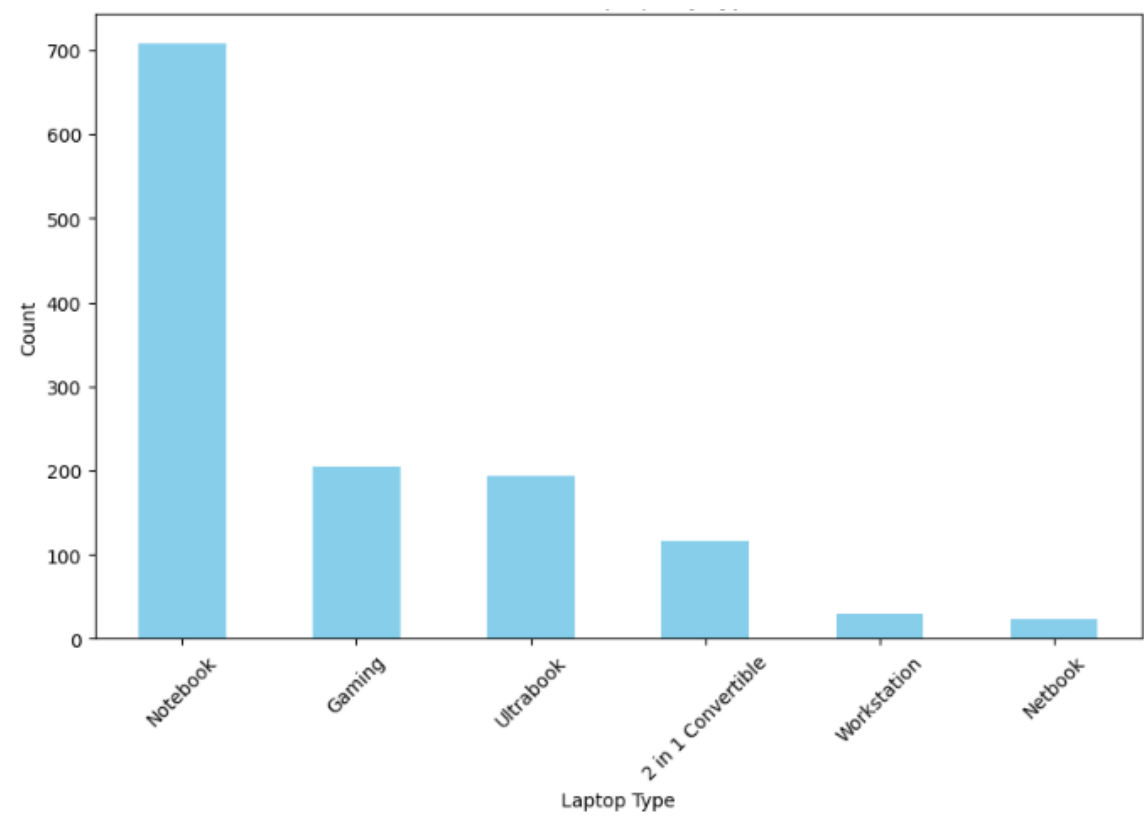
These metrics help us understand the spread and central tendency of each variable.

- **Histograms** and **box plots** were used to visualize the distribution and detect outliers in features such as:
 - RAM (GB)
 - Storage (GB)
 - CPU Frequency (GHz)
 - Weight (kg)
 - Price (Euro)

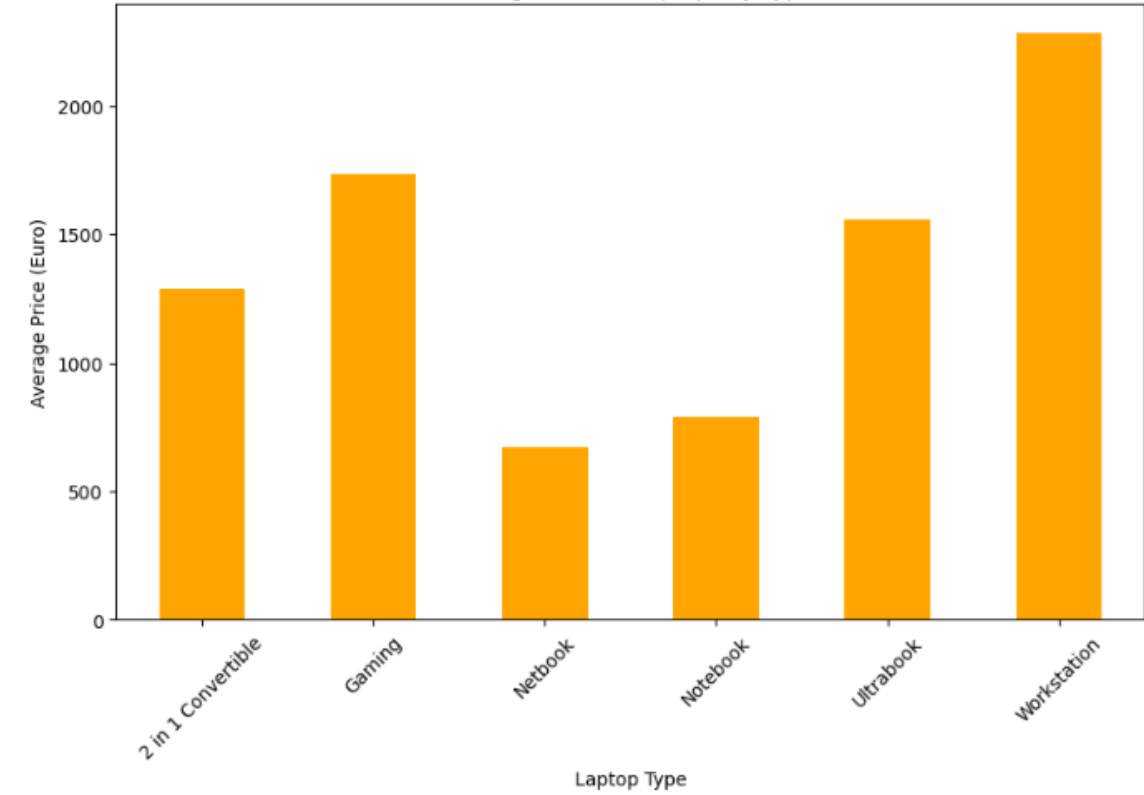
Correlation Heat Map



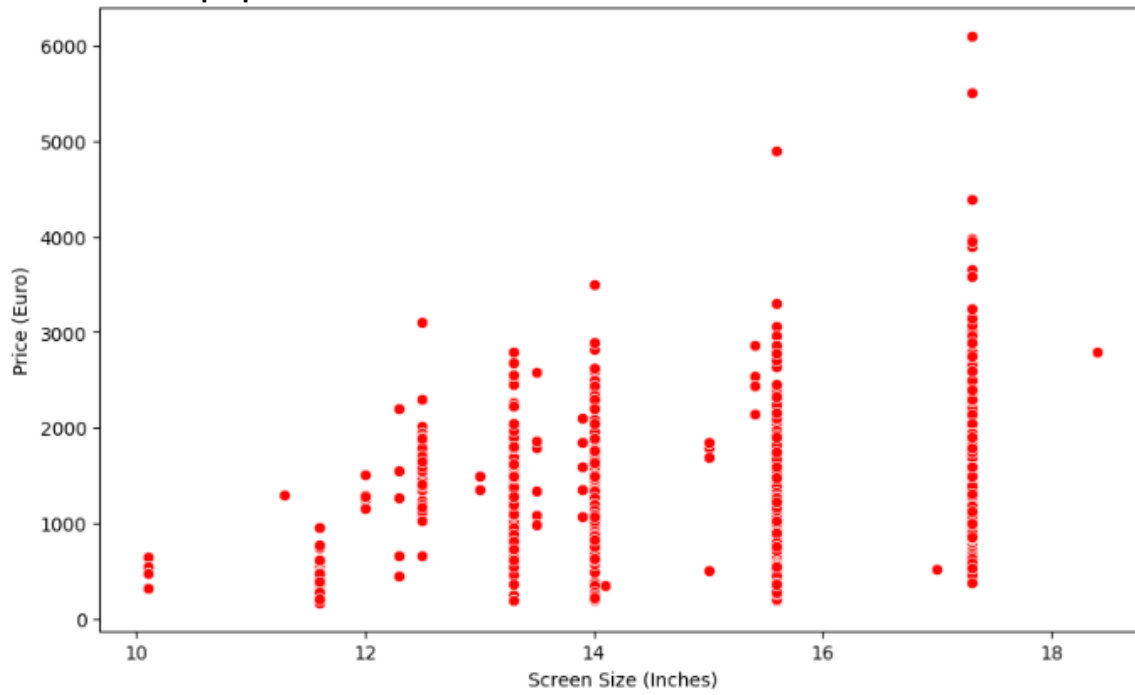
Bar Chart



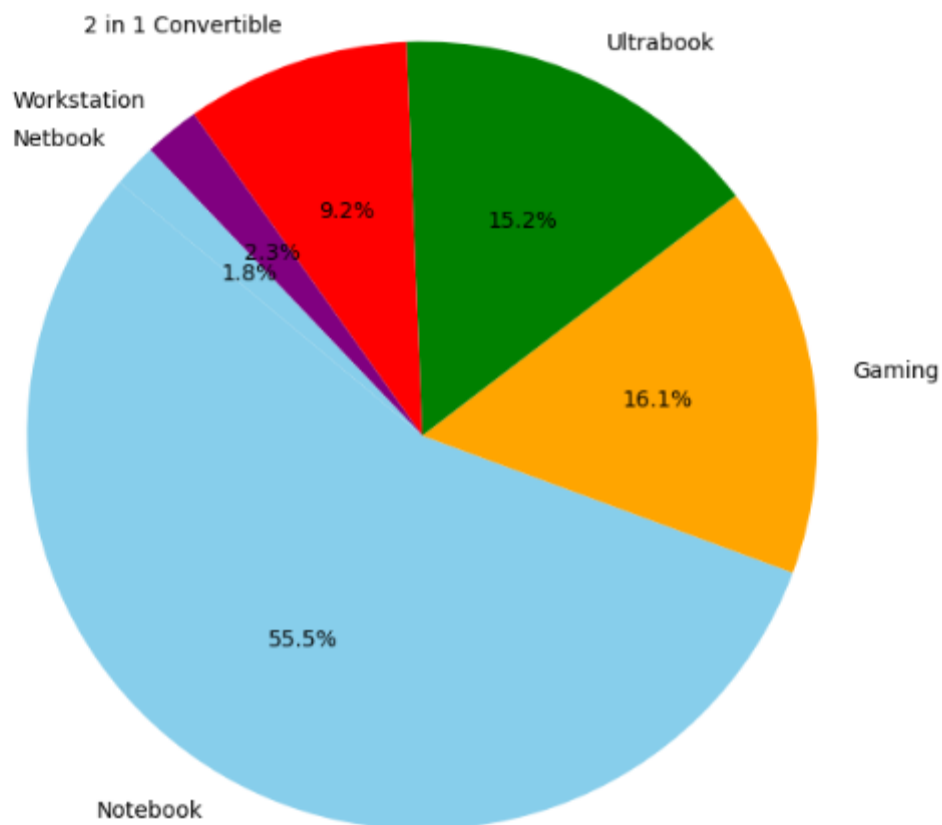
Average Laptop Prices by Category



Scatter Plot of Laptop Screen Size and Price:



Pie Chart:



Conclusion

This analysis reveals that laptop pricing is largely driven by technical specifications such as **RAM**, **CPU frequency**, and **screen size**. While **brand** and **operating system** also contribute to price variations, their influence is often more subjective and tied to consumer perception. The dataset proved to be clean, complete, and well-structured, making it suitable for further machine learning applications. Visualizations significantly enhanced our ability to interpret key trends, and transforming categorical variables into numerical format is a crucial step for building predictive models in the next phase of this project.